

Abubakar Khawaja

Senior Full-Stack Engineer | React | Next.js | TypeScript | Python | Django | AWS

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PROFESSIONAL SUMMARY

Full-Stack Engineer with 5+ years of experience architecting and shipping production software across Python and Django, React, TypeScript, and PostgreSQL. I take features from first idea to live deployment, prioritizing clean UI architecture, highly concurrent APIs, and robust asynchronous pipelines that hold up under real-world load. I have run production services on AWS with Kubernetes and EKS, PostgreSQL on RDS, and Redis caching, and I have built agentic AI features with tool calling and vector retrieval. I am comfortable owning system design, database performance, and critical third-party integrations end to end.

EXPERIENCE

Senior Full-Stack Engineer | Arbisoft

2021 – 2026

- Architected asynchronous processing pipelines using Celery and Redis for bulk document generation, scheduled jobs, and reporting systems managing 1,000+ assets.
- Owned end-to-end third-party integrations across multiple production platforms, including Stripe for payments, SendGrid for transactional emails, and Looker for BI analytics.
- Cut API response times by 30-70% across core endpoints by resolving N+1 queries, optimizing ORM access patterns, and implementing strategic PostgreSQL indexing.
- Redesigned React frontend architecture, reducing unnecessary re-renders by 30% and improving page load times by 20-30% via memoization, code-splitting, and optimized state management.
- Designed secure REST APIs with Django REST Framework, implementing JWT authentication, authorization, and third-party integrations.
- Implemented SSR, SSG, and ISR using Next.js while improving Core Web Vitals, accessibility, SEO, and overall application responsiveness.
- Elevated automated test coverage from 10% to over 60% (Jest, Cypress, Vitest) and integrated quality gates (ESLint, Husky), reducing production bugs by 25%.
- Reworked GitLab CI/CD pipelines with parallel jobs and layer caching, reducing deployment times by 40%.
- Deployed and tuned services on AWS ECS/EKS and Lambda to support scaling, build performance, and deployment reliability.
- Owned Sentry monitoring to cut production error volume: error triage with session replay, release and performance tracking, Slack and GitHub integration, and PII-safe payload encryption that keeps sensitive data out of captured events.
- Used AI coding tools such as Cursor, Claude Code, Codex, GitHub Copilot, and ChatGPT for both autonomous multi-step tasks and hands-on pair-programming, applying them to PR review, architecture discussions, and custom internal tooling, while comparing model outputs using structured prompts.
- Mentored junior engineers through regular code reviews and drove Agile feature delivery in tight collaboration with UX/UI designers and backend teams.

SELECTED PROJECTS

Telehealth Platform · [Django](#) + [PostgreSQL](#) + [Kubernetes](#)

- Architected and maintained a secure, HIPAA-compliant telehealth platform using Django, PostgreSQL on AWS RDS, and Kubernetes for container orchestration.
- Optimized patient-provider matching workflows through SQL query tuning and Redis caching, collaborating with data engineering to streamline ETL pipelines for real-time data processing.
- Integrated HL7 ADT (admit, discharge, transfer) message processing so patient events sync between clinical systems as they happen, with HIPAA-compliant handling of PHI in transit.

FixAlert, Maintenance Platform · [React](#) + [Django](#)

- Implemented Celery-based background processing, Stripe payment integration, and scalable reporting services.
- Optimized backend performance by eliminating N+1 queries and improving database access patterns.
- Designed asynchronous Celery workflows for QR generation, bulk PDF exports, and automated report generation.

Landit, Career Platform · [React](#) + [Next.js](#) + [TypeScript](#)

- Built pages with React Server Components on the Next.js App Router, keeping data fetching on the server and reserving client components for the interactive parts of the page.
- Executed zero-downtime frontend framework migrations (Material UI v4 to v5, React Router v4 to v5).
- Developed custom Looker BI visualizations and scalable SendGrid email templates.

Retrieval Chat, Agentic AI Assistant · [Vercel AI SDK](#) + [OpenAI](#) + [Supabase pgvector](#) · [Personal Project](#)

- Built a streaming chat interface with the Vercel AI SDK, rendering model output token-by-token with markdown formatting and sanitised HTML.
- Implemented agentic tool calling where the model selects between a custom pgvector knowledge-base retriever and live web search, surfacing the chosen source alongside the answer.

CORE SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, SQL, HTML5, CSS3/SCSS

Frontend: React.js, Next.js (App Router, Pages Router), Vue.js, TypeScript, Redux Toolkit, TanStack Query, React Hook Form, Tailwind CSS, Material UI, SCSS

Backend: Python (Django, Django REST Framework, FastAPI, Flask), Node.js (NestJS, Express), REST API design, GraphQL, webhooks, JWT & OAuth2 authentication, Celery & RabbitMQ (async jobs), WebSockets, Scrappy

Databases: PostgreSQL, MySQL, MongoDB, Redis, Prisma, Drizzle ORM, Django ORM, Database Indexing, Query Optimization, Caching

Cloud & DevOps: AWS (ECS, EKS, Lambda, S3, RDS, Cognito, API Gateway), Docker, Kubernetes, GitHub Actions, GitLab CI/CD, Datadog, Sentry, Nginx, Linux, Git

Testing: Jest, Cypress, Vitest, React Testing Library, Pytest, Django Test Framework, Testcontainers, ESLint, Prettier, Husky

Integrations: Stripe (payments), SendGrid & Resend (email), Firebase Auth, Google Maps, PostHog & Google Analytics, Google Looker (BI), JWT/OAuth2

AI/ML: LLM Application Architecture, AI Agents, Tool/Function Calling, Structured Outputs (Zod, Pydantic), Model Context Protocol (MCP), RAG, Embeddings & Vector Databases (pgvector), Prompt & Context Engineering, OpenAI / OpenRouter / Ollama (local models), Vercel AI SDK

AI Development: ChatGPT, GitHub Copilot, Cursor, Claude Code, Codex, Prompt Engineering, AI-Assisted Development

Foundations: Data Structures & Algorithms, System Design, Performance Optimization, Agile/Scrum, Code Review, Mentoring

EDUCATION

B.S. Software Engineering Punjab University College of Information & Technology (PUCIT) · GPA: 3.08

CERTIFICATIONS

- The AI Engineer Path & Prompt Engineering for Web Developers — Scrimba (July 2026)